



NA.YADRA Project Capability Summary

Pacific Maritime Intelligence Platform

This document summarises the current capability of the NA.YADRA maritime intelligence dashboard from its early build stage to its present operational prototype.

1. Current Project Capability

Capability	What it does now
Live AIS vessel tracking	Displays vessels around Fiji and the wider Pacific using AISStream data.
Risk scoring	Scores vessels as LOW, MEDIUM, or HIGH based on behaviour patterns.
Behavioural detection	Detects night movement, offshore low speed/stopped vessels, possible night stop locations, repeat activity, and stopped outside port zones.
Port proximity logic	Compares vessel location against known port zones such as Lautoka/Vuda, Denarau/Nadi, and Suva Harbour.
Dashboard map	Shows vessels on an interactive map with satellite and street-map style overlays.
Intel summary panel	Summarises total vessels, high-risk vessels, medium-risk vessels, low-risk vessels, and stale contacts.
Cyber feed health panel	Tracks cyber feed status such as OpenPhish integration and last check time.
Night movement report	Generates PDF intelligence reports for vessels detected moving during Fiji night hours.
Possible night stop report	Generates PDF reports for vessels stopped or slow offshore during Fiji night hours.
Analyst notes	Allows suspicious vessel behaviour to be written up as intelligence leads.

2. Dashboard Panels Explained

Panel	Purpose
Map Panel	Main operational view showing vessel locations, port zones, alert markers, and movement patterns.
Vessel Cards	Lists vessels with risk level, score, offender status, speed, and behavioural flags.
Intel Summary	Gives a quick operational snapshot of vessel risk across the monitored area.
Cyber Feed Health	Shows whether cyber threat feeds are active and when they were last checked.
Alert / Behaviour Popups	Explains why a vessel was flagged, including MMSI, score, nearest port, distance, coordinates, and behaviour type.
PDF Report Output	Creates exportable reports suitable for briefing, analyst review, or sharing with trusted maritime contacts.

3. Operational Value Today

NA.YADRA can already act as a prototype maritime intelligence support tool. It can highlight vessels behaving unusually near Fiji waters, especially at night, near port approaches, or outside normal port zones. It does not replace official systems, but it can provide early warning leads, visual intelligence summaries, and structured reports for analyst review.

4. Future Capability With Stronger Satellite Fusion

With stronger satellite capability and Innovate UK support, NA.YADRA could add higher-resolution imagery, more frequent satellite refresh, AIS-dark detection, satellite-AIS mismatch detection, small craft/mothership detection, automated daily intelligence briefings, secure cloud deployment, and wider Pacific coverage.

Important: All detections are intelligence leads for review only. A flagged vessel does not confirm criminal activity.